

Clinical Risk Assessment Report: Patient 73

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 9.5% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.43x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.5%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.55 [Avg (27th %ile)] (-2.5%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Carcinoembryonic Antigen (CEA) **Value: 1.8 [Bottom 20%] (-1.0%)**

Elevated levels are strongly associated with increased tumor burden and potential for micrometastasis.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: 0.12 [Top 17%] (-1.0%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.65 [Bottom 18%] (-0.9%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -1.91 [Bottom 4%] (-0.6%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

Patient Age (Age) **Value: 60.0 [Avg (35th %ile)] (-0.6%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: 0.04 [Top 17%] (-0.6%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: -0.83 [Bottom 18%] (-0.5%)**

Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

Machine Learning Feature Attribution (SHAP Decision Path)

